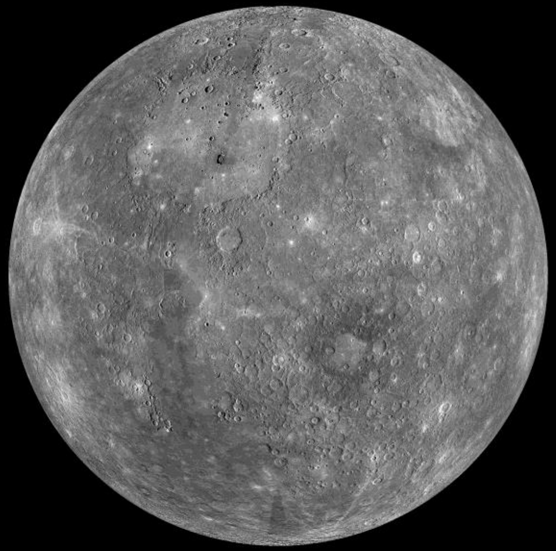
Why Earth Isn’t a Frozen Rock or a Boiling Cauldron?

By: Valentina Sánchez-Castaño

Why do you think this is the temperature of these planets?

Mercury



333 °F (167 °C)

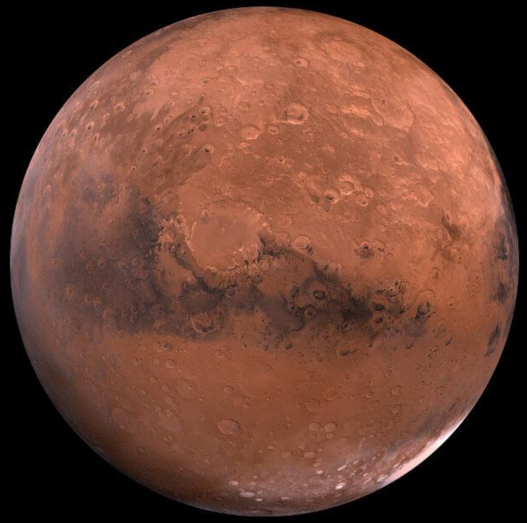
Venus

867 °F (464 °C)

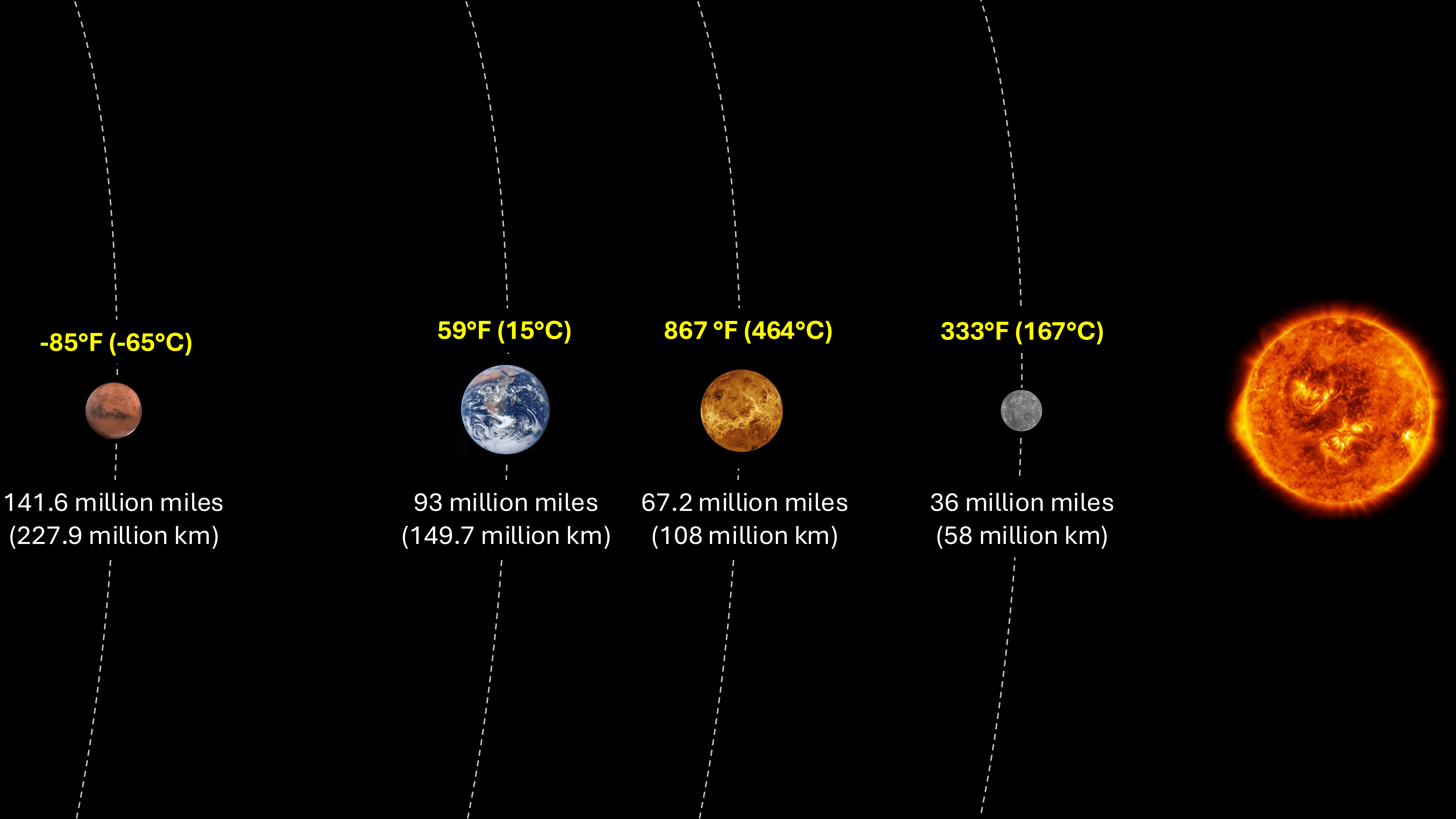
Earth

59 °F (15 °C)

Mars



-85 °F (-65 °C)


-85°F (-65°C)
141.6 million miles
(227.9 million km)
59°F (15°C)
93 million miles
(149.7 million km)
867 °F (464°C)
67.2 million miles
(108 million km)
333°F (167°C)
36 million miles
(58 million km)

Learning Goals: Decoding the Climate

By the end of this session, you will be able to:

Deconstruct the Energy Budget

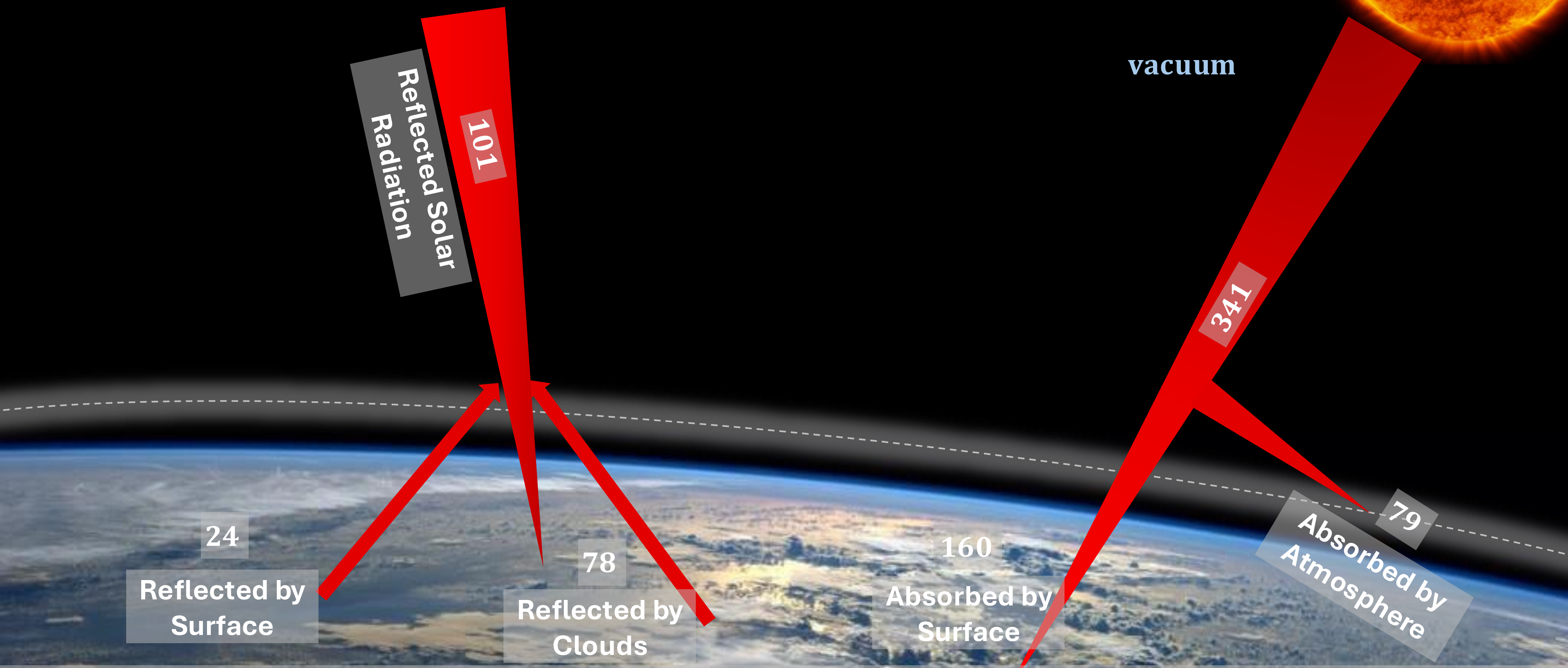
Identify Heat Transfer Methods
(Radiation, **Conduction**, and **Convection)**

Master the Concept of Albedo

Solve the Planetary Mystery

Connect Science to El Paso

Energy In vs. Energy Out
First part – Shortwave



Energy In vs. Energy Out
Second part – Longwave

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Longwave Radiation

Greenhouse effect:

This process acts as a natural "blanket" for the planet:

The Process: Instead of escaping directly into space, much of Earth's infrared radiation is absorbed by the atmosphere and clouds, then sent back toward the surface.

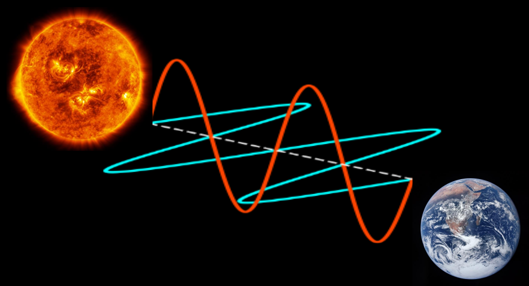
The Benefit: This natural warming keeps Earth above the freezing point of water, making life possible.

The Human Factor: By burning fossil fuels and clearing forests, humans have intensified this effect. This trapped heat has moved beyond natural levels, resulting in **global warming**.

How Does Heat Actually Move?

Radiation

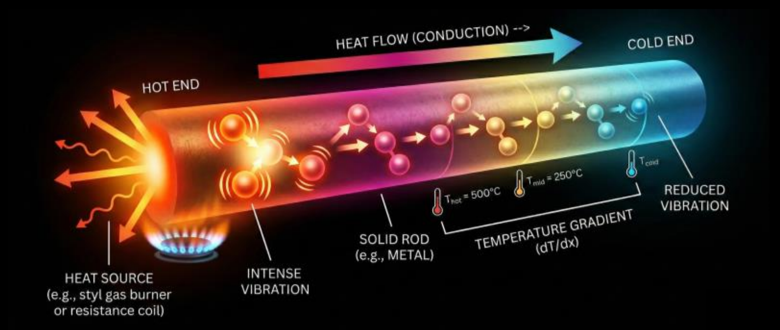
The "Space" Traveler



It doesn't need a medium (like air or water) to travel. This is the **only** way the Sun's energy reaches Earth through the vacuum of space.

Conduction

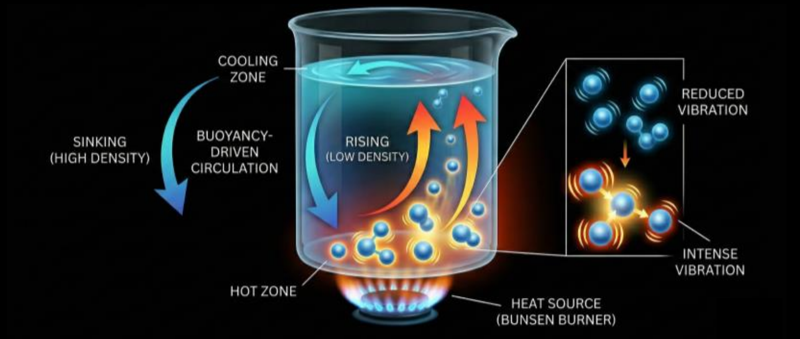
The "Touch" Transfer



Heat moving through direct contact.

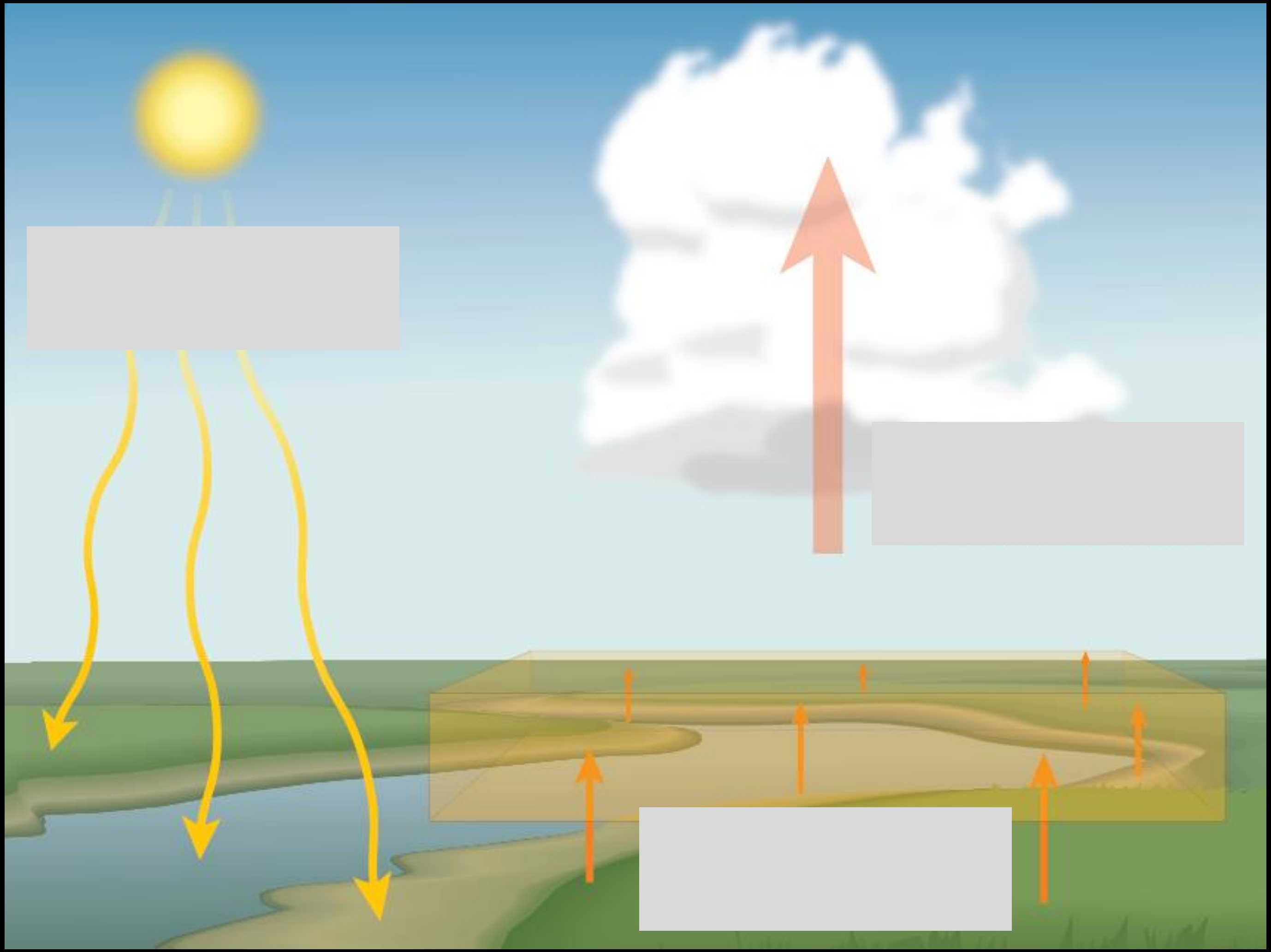
Convection

The "Fluid" Engine



Heat moving through the circulation of liquids or gases.

How Does Heat Actually Move?

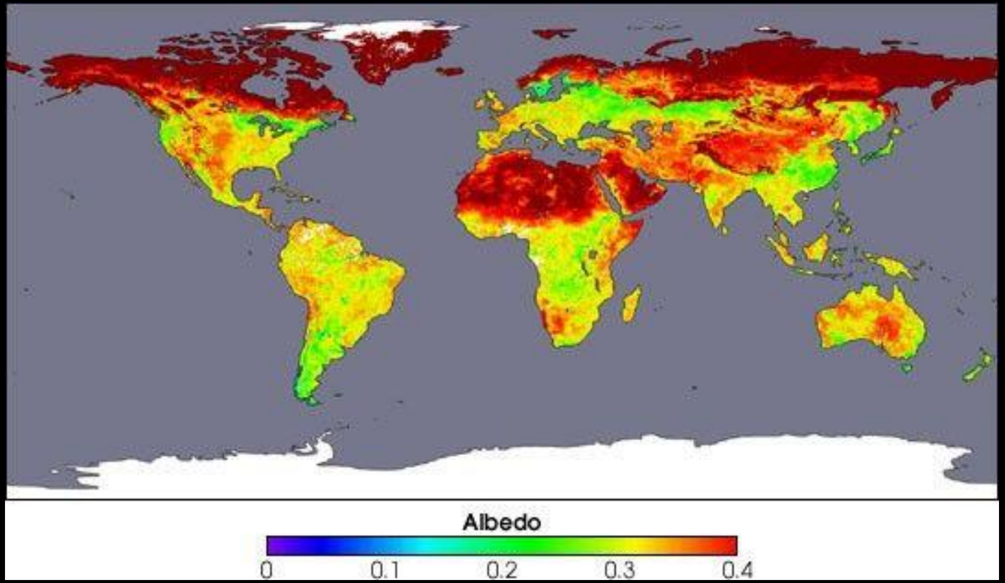


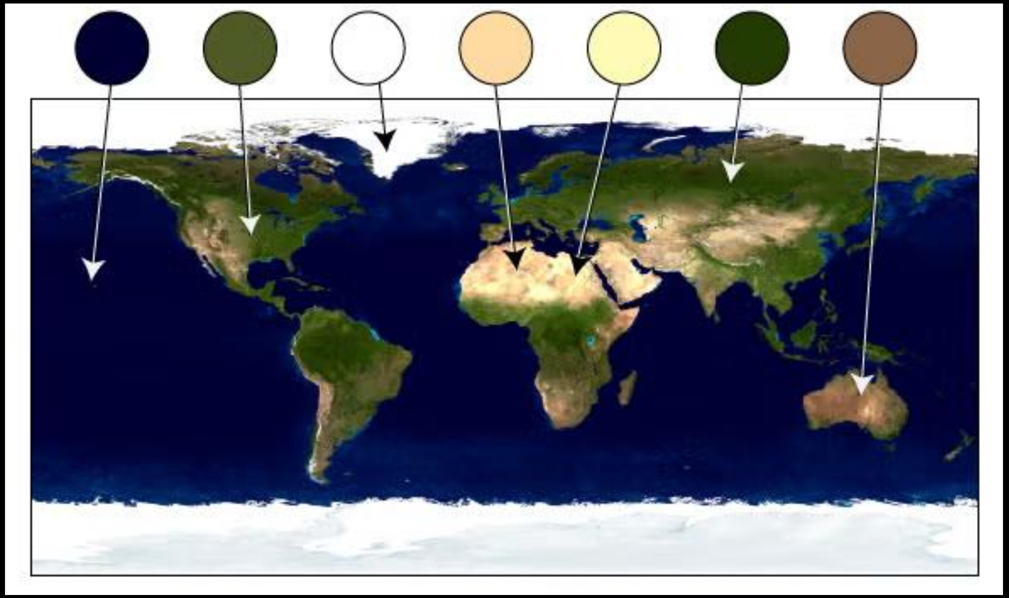
What is albedo?

Albedo is a measure of how much light a surface reflects back rather than absorbing.

It is measured on a scale from **0** (total absorption/black) to **1** (total reflection/white)

The albedo of the Earth is 0.294.



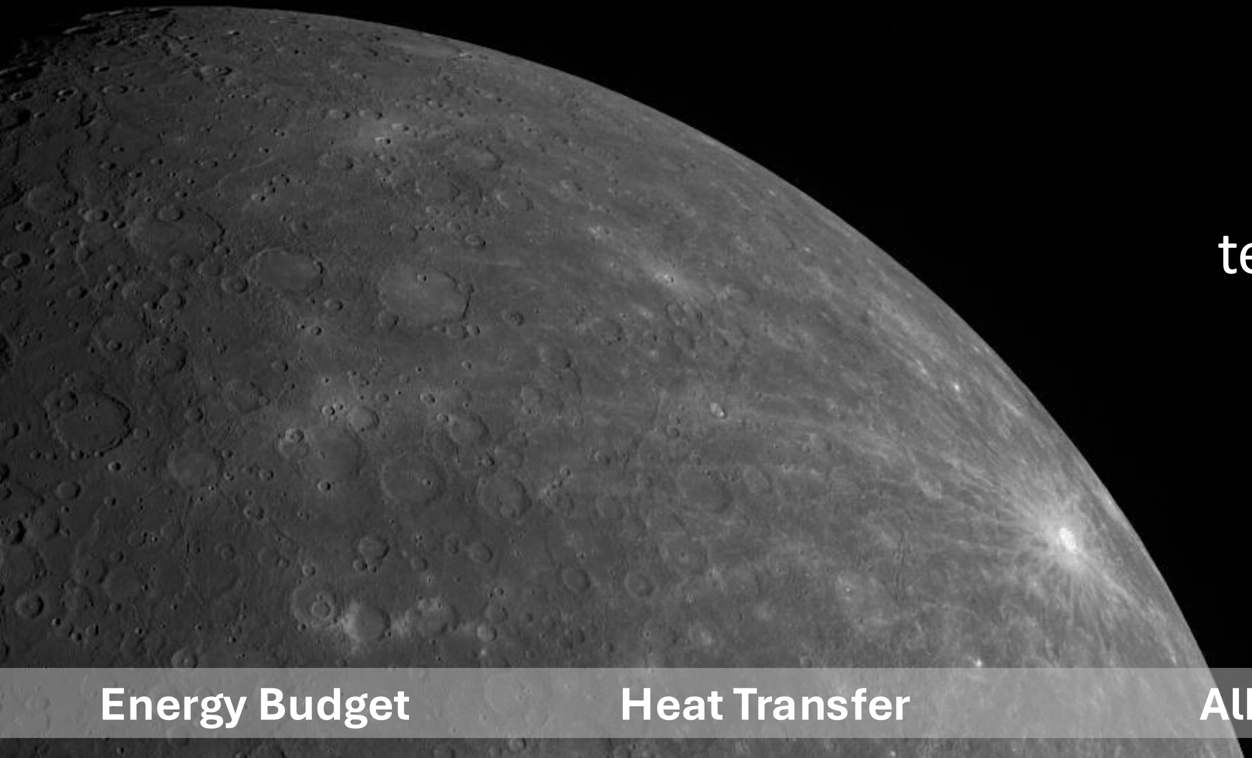


The Venus Temperature

Mercury is close to the Sun (167 °C), but Venus is hotter (464 °C). Why?

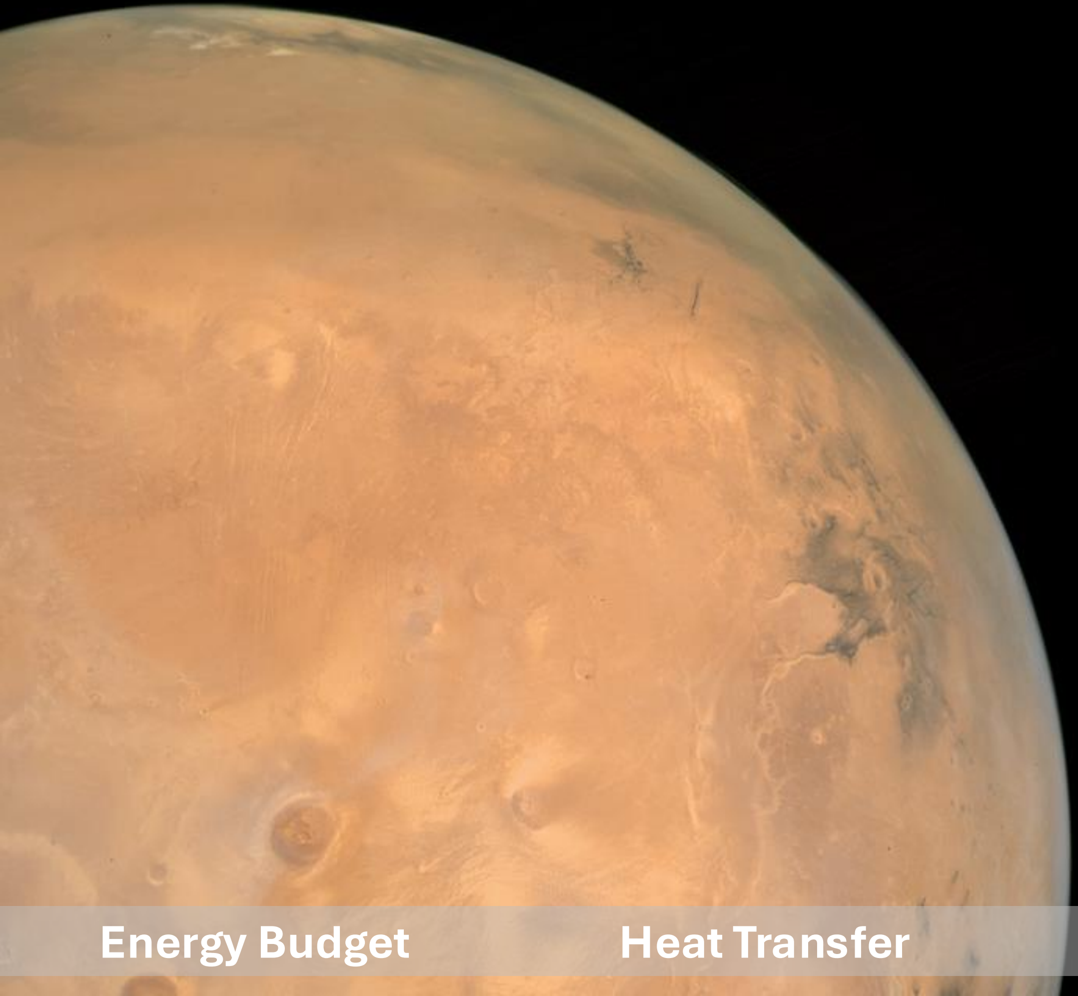
The Venus vs. Mercury Showdown

Mercury is close to the Sun (167 °C), but Venus is hotter (464 °C). Why?



Mercury's surface temperatures are both extremely hot and cold. Because the planet is so close to the Sun, day temperatures can reach highs of 800°F (430°C). Without an atmosphere to retain that heat at night, temperatures can dip as low as -290°F (-180°C).

What about Mars?



While Earth has a thick enough atmosphere to keep our temperatures relatively stable, Mars has an extremely thin atmosphere, about **100 times thinner** than Earth's. Because of this "thin blanket," Mars cannot hold onto heat.

The El Paso Heat Detective Challenge

The Scenario:

It is 3:00 PM on a Tuesday in July. The sun is blazing over the Franklin Mountains. We are measuring the ground temperature in two different spots.

Clue 1: The Surface. Spot A is a freshly paved black asphalt parking lot in Downtown. Spot B is the sandy, light-brown dirt in a natural trail at Franklin Mountains State Park.

Clue 2: The Albedo. One spot has an Albedo of **0.05**. The other has an Albedo of **0.40**.

Clue 3: The Infrared "Breeze." If you hold your hand 1 inch above the surface at Spot A, it feels like an oven. At Spot B, it feels warm, but not painful.

Clue 4: The Nighttime Mystery. At midnight, Spot A is still releasing heat into the air. Spot B is already much cooler.

The El Paso Heat Detective Challenge

The Challenge:

1. **Identify:** Which spot has the **lower** Albedo? Why?

2. **Explain:** What type of heat transfer (**Radiation, Conduction, or Convection**) are you feeling when you hold your hand above the asphalt?

3. **Predict:** If we painted all the Downtown parking lots the same color as the desert sand, what would happen to the temperature of the city?

Thank you